

Impact of Pokémon Go on Physical Activity:
A Systematic Review and Meta-Analysis

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Context: Pokémon Go is a popular mobile augmented reality game that requires players to travel to different locations to capture virtual characters. This study systematically reviews and quantifies Pokémon Go in relation to physical activity engagement among players.

Evidence acquisition: A keyword search was conducted in PubMed, Web of Science, Scopus, EBSCO, SPORTDiscus, PsycINFO, ScienceDirect, and Cochrane Library for articles published between July 2016 and October 2018. Meta-analysis was performed to estimate the pooled effect of playing Pokémon Go on physical activity outcome.

Evidence synthesis: From the keyword search, 17 studies (16 observational and 1 pre–post) were identified, with a total sample of 33,108 participants. A comparison between Pokémon Go players and nonplayers and between pre- and post-play time points revealed an increase in walking duration, distance walked, and number of steps/day. Pokémon Go players were also found to engage in less sedentary behavior. Playing Pokémon Go was associated with an increase in the number of steps per day by 1,446 steps (95% CI=953, 1,939; $I^2=81\%$).

Conclusions: Playing Pokémon Go was associated with a statistically significant but clinically modest increase in the number of daily steps taken among game players. One challenge for future physical activity interventions using Pokémon Go is to retain active engagement once the initial novelty wears off. Additional studies with longer follow-up periods and experimental study design are needed to assess to what extent Pokémon Go and other augmented reality games can be used to promote physical activity at the population level for a sustained time period.

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CONTEXT

Being physically active can help prevent chronic disease, maintain a healthy weight, and improve quality of life.¹ The rapidly growing availability and use of mobile health applications creates opportunities to promote physical activity and an active lifestyle at the population level.² Pokémon Go was not designed to explicitly promote physical activity, yet this game has had great success in reaching millions of people and holds the potential to influence daily activities. Before Pokémon Go, other exergames (i.e., the new generation of active video games) that incorporate aspects of physical activity have gained popularity.³ Previous research shows that playing exergames may promote energy expenditure, weight loss, and overall health.⁴ Researchers have previously examined

the potential of GPS-based games that promote physical activity via interactive gameplay, such as geo-caching.⁵ These types of games present an enjoyable and appealing alternative to traditional physical activities⁶ and offer opportunities for increased social interaction and time spent outdoors.^{7,8} However, public health efforts to encourage physical activity via a large number of these

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exergames or other types of activity-based video games are yet to be implemented, given that most of them have mixed evidence.⁹

The new era of digital technology use, such as smartphones with a wide range of health applications, provides an opportunity to utilize these tools as interventions to help people increase their physical activity level. Despite the seemingly high popularity of these applications, the evidence for their ability to promote and sustain individuals' physical activity and other health habits (e.g., adherence to healthy sleep schedules) remains limited.¹⁰ Some of the physical activity applications in existence today provide users with enhanced delivery of behavioral change strategies. Specifically, some modern mobile applications support personalization, which makes it easier to self-monitor behavior progress, track changes, set goals, use reminders, personalize information, and engage with the interactive mobile interface at any time.¹¹ These features may increase an individual's motivation and reinforce changes in behavior.

Traditional face-to-face interventions that are delivered on a personal level tend to be more expensive and have a limited reach.¹² By contrast, the rapid growth of mobile phone applications and their direct mode of intervention delivery can reach more people and potentially influence their health behavior in terms of changes in the level of physical activity and time spent sedentary and their consequent impact on health.¹³ Of note, some epidemiological evidence has shown that moderate-to-vigorous physical activity reduces risks of spending time in sedentary behavior, whereas substantial time spent in sedentary behavior negatively impacts cardiometabolic well-being.¹⁴

One of the more recent examples of mobile applications with attempt to increase time spent in physical activity and reduce sedentary behavior is the Pokémon Go mobile augmented reality game, which rapidly acquired 44 million users within several weeks of its debut in the summer of 2016.¹⁵ The goal of the game is to catch as many special virtual characters as possible, requiring users to walk to various locations, using the mobile phone's GPS.¹⁶ Although Pokémon Go has been released only recently, studies have reported mixed findings on its health impact. This study helps fill the gap in the literature by conducting the first systematic review and meta-analysis on the impact of Pokémon Go on physical activity. The review aims to identify and critically appraise physical activity outcomes in relation to Pokémon Go. It is hypothesized that playing Pokémon Go is associated with an increased level of physical activity among users of all ages.

EVIDENCE ACQUISITION

This systematic review and meta-analysis was conducted in accordance to PRISMA guidelines.¹⁷

Studies that met all the following criteria were included in the review: (1) study design, experimental or observational study; (2) population, human participant of any age; (3) exposure, use of Pokémon Go; (4) outcome, physical activity; (5) article type, peer-reviewed publication of an empirical original study; (6) time window of search, from July 2016 when Pokémon Go was first released to October 2018; and (7) language, article written in English.

A systematic search was performed in 8 electronic bibliographical databases: PubMed, Web of Science, Scopus, EBSCO, SPORTDiscus, PsycINFO, ScienceDirect, and Cochrane Library. Given the small review scope, the research team decided to use a single keyword, *Pokémon Go*, which enabled identification of all potentially relevant studies. Titles and abstracts of the articles identified through the keyword search were screened against the study selection criteria. Retrieved articles were then screened for eligibility and selected for inclusion for the full-text evaluation. Studies were independently, blindly screened and selected by 2 co-authors. Discrepancies were resolved through discussion. Inter-rater agreement was assessed using Cohen's κ ($\kappa=0.73$).

Data were systematically extracted using a predetermined standardized form. The following information was extracted from each included study: author(s), publication year, study design, sample size, sample characteristics such as age and sex distribution, intervention arms, intervention frequency and duration, adjustment for baseline characteristics, key findings and estimated effects on physical activity, and statistical approach.

Meta-analysis was performed to estimate the pooled effect size of Pokémon Go on physical activity, measured by changes in daily number of steps taken ($n=4$; difference in means in steps/day). Pooling of walking ($n=7$), jogging ($n=1$), running ($n=2$), skating ($n=1$), or biking ($n=1$) duration (minutes/week); exercise duration and frequency including moderate-to-vigorous- and light-intensity activity ($n=7$; minutes/week); distance walked ($n=2$; miles/week); and sedentary behavior ($n=5$; minutes/day) was infeasible given the small number of studies that assessed each of these measures. The treatment arm was defined as a group that received or played Pokémon Go, whereas the control arm was a group that did not receive or play Pokémon Go. Some studies also introduced a third arm called an ex-players group, defined as those who had previously played but were not currently playing Pokémon Go.

Study heterogeneity was assessed using the I^2 index. A fixed effects model was estimated when $I^2 \leq 50\%$, and a random effects model was estimated when $I^2 > 50\%$. Publication bias was assessed by visual inspection of the funnel plot and the Begg's and Egger's tests. All statistical analyses were conducted using Stata, version 15.1.

The study quality assessment was based on the Quality Assessment Tool for Quantitative Studies developed by the Effective Public Health Practice Project¹⁸ and included the following: study design, data collection methods, confounders and selection bias, and withdrawals and dropouts. A global rating for a study was obtained based on the rating of each quality component being

either strong, moderate, or weak. The strong overall rating corresponds to the absence of weak components, moderate rating corresponds to at least 1 weak component, and a weak rating includes 2 or more weak components. The detailed description of the quality assessment tool is available elsewhere.¹⁹ Study quality assessment helped measure the strength of scientific evidence but was not used to determine the inclusion of studies.

EVIDENCE SYNTHESIS

Figure 1 illustrates the PRISMA flow chart of the screening process. The database search yielded 269 articles, including 9 from PubMed, 87 from Web of Science, 90 from Scopus, 27 from EBSCO, 5 from SPORTDiscus, 29 from PsycINFO, and 22 from ScienceDirect. In addition, 4 articles were identified through a hand search. After removing duplicates, 132 articles remained, of which 103 were excluded based upon a screening of the title and abstract. The remaining 29 full-text articles were screened against the study criteria. Of these, 12 articles were excluded for the following reasons: physical activity as an outcome was not included in 11 articles,^{7,20–29} and 2 articles included the same study participants^{15,29} so 1 article was excluded.¹⁵ In total, 17 articles were included in the review.^{3,6,9,15,16,29–40}

Study and participant characteristics are summarized in Table 1. Sample sizes varied across studies, with a range from 26 to 27,126 participants. Three studies included <100 participants, and the remaining studies ($n=14$) had a sample size >100 participants, with 1 study including >20,000 participants. The 16 observational^{6,9,15,16,29–40} studies included 33,745 participants and the only pre–post study³ included 167 participants. The total sample of the included studies was 33,108 participants who were either Pokémon Go players, nonplayers, or ex-players. The mean age across studies ranged from <10 to 60 years, with most participants being male (approximately 70%). One of the possible explanations is that designers of Pokémon Go observed that the game is more appealing to male than to female players.⁹ Two studies included children and adolescents (with the mean age of 11 years), whereas the remaining studies exclusively included adults aged ≥ 18 years.

All studies were published between 2016 and 2018. These studies were conducted in the U.S. ($n=10$), Spain ($n=2$), United Kingdom ($n=1$), Thailand ($n=1$), Costa Rica ($n=1$), and Hong Kong ($n=2$). Intervention duration varied, ranging from 60 minutes to 10 weeks in length. Physical activity was assessed subjectively in 13 studies, using questionnaires such as the International Physical Activity Questionnaire, Physical Activity Readiness Questionnaire, Godin–Shephard Leisure Time Physical Activity Questionnaire, and the Adolescent Physical Activity Measure Questionnaire. Seven studies

incorporated different objective assessments of physical activity, including accelerometers, pedometers, wrist-worn activity trackers, analog electrocardiogram heart rate monitors, and smartphone applications.

Table 2 summarizes the estimated effect of Pokémon Go on physical activity outcomes, statistical methods, and participants' baseline characteristics. Physical activity outcomes were assessed by the number of steps taken ($n=7$),^{3,9,15,32,33,35,36} duration of time spent being physically active ($n=10$),^{6,16,29,30,32,34,35,38–40} time spent engaging in sedentary behavior ($n=5$),^{6,30,32,38,40} walking distance ($n=3$),^{16,36,37} and energy expenditure ($n=2$).^{29,38} The studies controlled for baseline characteristics including age, sex, race/ethnicity, annual household income, body weight, BMI, and prior physical activity level.

A comparison between Pokémon Go players and nonplayers^{16,29,30} and a comparison between pre- and post-play time points^{6,31,34,35} revealed an increase in walking duration (minutes/day), distance walked (miles/week), and number of steps/day.^{3,9,33,36} In addition, 2 studies investigated the effect of Pokémon Go play time on time spent engaging in sedentary behaviors and concluded that players were less likely to spend time sitting when playing Pokémon Go.^{6,38}

Figure 2 is the forest plot based upon the meta-analysis on the influence of Pokémon Go on number of steps/day. A total of 6 studies^{3,9,15,33,35,36} reported the number of steps taken because of playing Pokémon Go. Among them, 2 were excluded from the meta-analysis^{35,36} owing to insufficient data reported on estimation uncertainty (e.g., SEs and CIs). The remaining 4 studies with a total of 2,692 participants were included in the meta-analysis.^{3,9,15,33} Playing Pokémon Go was found to be associated with an increase in the number of steps/day by 1,446 steps (95% CI=953, 1,939; $I^2=81\%$; random effect model). No publication bias was indicated by the Begg's ($p=0.73$) and Egger's ($p=0.43$) tests.

Most of the studies were rated as strong in data collection methods, with only 3 that were rated weak.^{34–36} All studies were moderate in study design, selection bias, and addressing confounders. There were 3 studies that were rated as weak in the selection bias criterion^{16,32,38}; all others were rated as moderate. Rating on blinding was low across all 17 studies, as most of the outcome assessors and study participants were aware of the exposure status and the research question, respectively. Not all studies reported the proportion of participants who agreed to participate. Overall, the global ranking resulted in roughly 40% ($n=7$)^{3,6,9,15,29,33,40} of the studies ranked as moderate and the remaining nearly 60% ($n=10$)^{16,30–32,34–39} ranked as weak, with no study receiving a strong ranking. The quality assessment of the studies included in the review is summarized in Appendix Table 1, available online.

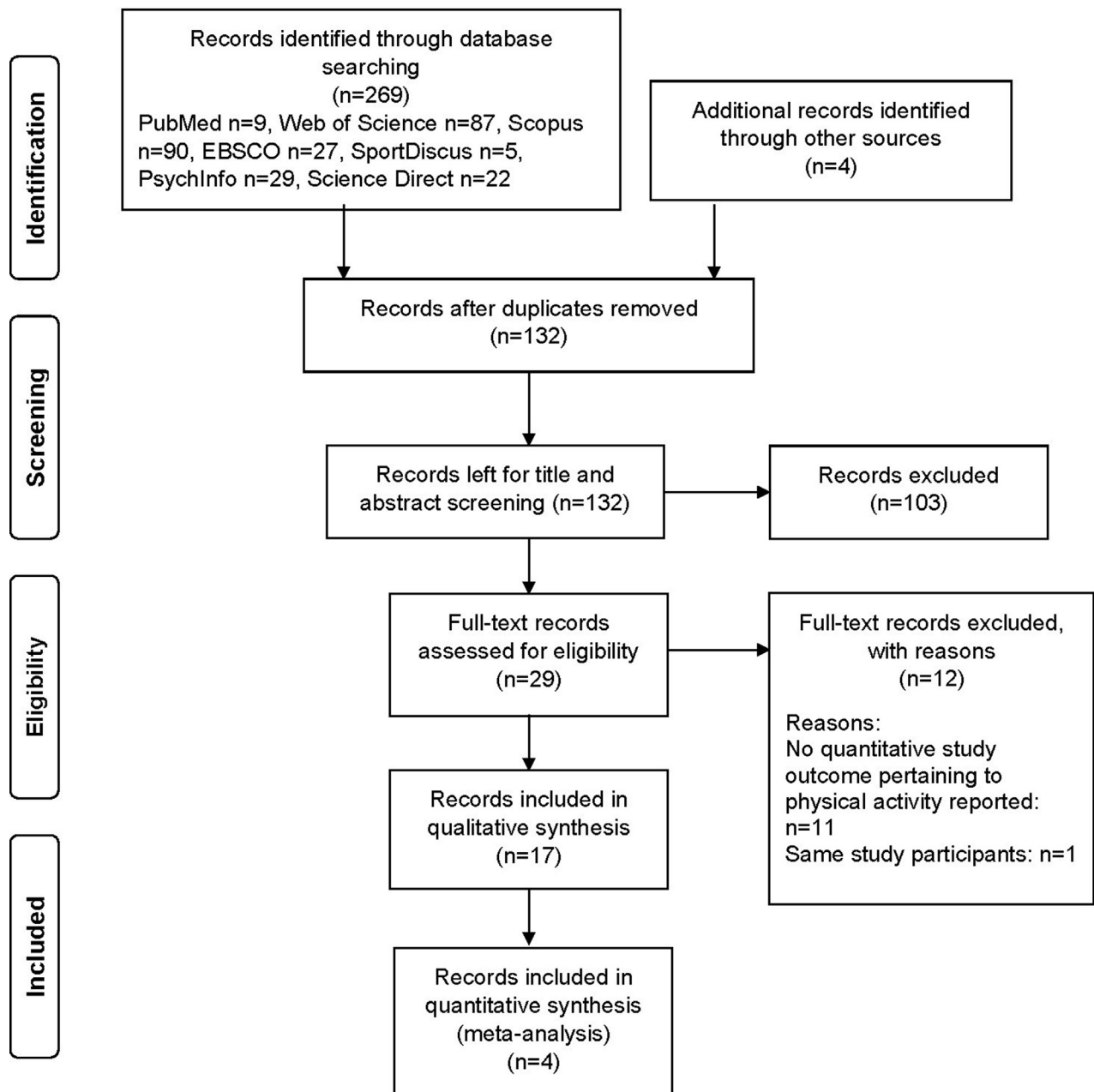


Figure 1. PRISMA study selection flowchart.

DISCUSSION

The primary focus of this review was to evaluate the impact of playing the Pokémon Go game on physical activity level. With rapidly evolving technologies in the field of behavioral medicine, Pokémon Go, though unintentionally, was able to reach masses of its users and influence their health behavior.⁴¹ Pokémon Go, not being a mobile health application, is an example of technology-leveraged behavioral intervention that was able to facilitate factors of physical activity outcomes. The

interface and functionality of the game exceeds traditional understanding of mobile health by providing a different way to individualize health interventions.^{39,40} Nevertheless, it should be noted that worldwide popularization of Pokémon Go raises concerns regarding potential risks players may encounter during their travel through various landscapes.¹⁵ In one study, 34% of the players reported some adverse events (e.g., bone fracture, trauma, and skin injury).²⁵ Another study reported that 9% of the sample suffered from accidents related to pedestrians, cars, or furniture.³¹ Player safety may need

Table 1. Basic Characteristics of the Studies Included in the Review

Study ID	First author (year)	Country	Study design	Sample size	Age (years)	Female (%)	Measures	Intervention duration
1	Althoff (2016) ⁹	U.S.	Observational	27,126	Median: 33	25	3 axis accelerometer; gyrometer of a wrist-worn consumer activity tracking fit-device -Microsoft Band	4 weeks
2	Barkley (2017) ⁶	U.S.	Observational	358	19.8±2.1	52	Personal physical activity monitor (e.g., accelerometers, pedometers), questionnaire (IPAQ)	3 weeks
3	Broom (2018) ³⁰	U.K	Observational	461	28.8±9.56	57	Short version questionnaire (IPAQ)	3 months
4	Escaravajal-Rodriguez (2018) ³¹	Spain	Observational	714	24.86±5.5	38	Ad-hoc opinion questionnaire	
5	Fontaine (2018) ³²	U.S.	Observational	27	21.5±2.6	70	Accelerometer; pedometer; analog electrocardiogram heart rate monitor and watch, PAR-Q questionnaire	60 minutes
6	Howe (2016) ³³	U.S.	Observational	1,182	26.49±4.55	71	iPhone 6 smartphone, questionnaire	10 weeks
7	Kogan (2017) ³⁴	U.S.	Observational	269	18 and older	68	Questionnaire	
8	Liu (2017) ³⁵	U.S.	Observational	47	28.7±5.9	47	Questionnaire	
9	Ma (2018) ³⁶	Hong Kong	Observational	210	26.1±8.7	33.8	iPhone “health” application, questionnaire	21 days
10	Madrigal (2018) ³⁷	Costa Rica	Observational	1,059	18 and older	52	National household survey	
11	Marquet (2018) ¹⁵	U.S.	Observational	74	19.6	50	Smartphone-enabled accelerometers, IPAQ short form	
12	Militello (2018) ³⁹	U.S.	Observational	160 adults, 31 children	N/A	71.9; 28.8	Revised Godin Leisure Time Exercise Questionnaire, family's characteristics and Pokémon GO user experience, healthy lifestyle beliefs scale, child characteristics and Pokémon Go user experience, parental influence on physical activity scale	
13	Nigg (2017) ⁴⁰	U.S.	Observational	N/A	28.6±8.5	57.8	Revised Godin leisure time exercise questionnaire	
14	Ruiz-Ariza (2018) ¹⁶	Spain	Observational	190	13.32±1.07	49	Adolescent Physical Activity measure questionnaire	8 weeks
15	Wattanapisit (2018) ³⁸	Thailand	Observational	26	22.04±1.70	26.9	Global Physical Activity Questionnaire	4 months
16	Wong (2017) ²⁹	Hong Kong	Observational	644	18–60	52	IPAQ short form	
17	Xian (2017) ³	U.S.	Pre–post	167	25 (median)	48	iPhone health application, questionnaire	

IPAQ, International Physical Activity Questionnaire; N/A, not applicable; PAR-Q, Physical Activity Readiness Questionnaire; UK, United Kingdom.

Table 2. Measures, Statistical Methods, and Outcome of the Studies Included in the Review

Author (year)	Physical activity outcome measure(s)	Components of arms	Intervention effect on physical activity outcome	Statistical analysis	Controls for baseline characteristics
Althoff (2016) ⁹	Daily steps (step/day)	2	Players: 5,756 (baseline) 5,948 (after 1 play) 7,229 (after 10 plays) 26% increase over 2 months (1,473 steps) Controls: 6,435 (baseline) 6,385 (after 1 play) 7,229 (after 10 plays) 13% increase over 2 months (794 steps)	Mann–Whitney <i>U</i> -tests	Age, sex, weight status (BMI), prior activity level
Barkley (2017) ⁶	Walking (minutes/day); sedentary behavior (minutes/day)	1	Baseline: 108.5±110.8 minutes/day Time 1: 218.6±156.3 minutes/day Time 2: 182.7±172.1 minutes/day 102% and 68% increase baseline: 346.6±201.3 minutes/day Time 1: 261.7 ±172.4 minutes/day Time 2: 284.3 ±175.4 minutes/day 25% and 18% decrease	ANOVA, χ^2	
Broom (2018) ³⁰	Vigorous, moderate physical activity (days, minutes); walking (days, minutes); sitting (days, minutes)	2 Players, controls	Players: Vigorous (minutes) 64.24±74.14 at Baseline and 42.17±46.61 at follow up moderate (minutes) 69.45±117.98 vs 61.74±83.22 Walk (minutes) 111.97±117.47 vs 100±91.7 sitting weekend (minutes) 386.04±200.50 vs 446.09±174.06 sitting weekdays (minutes) 328.33±188.92 vs 336.87±176.32 Controls: Vigorous (minutes) 69.03±67.68 vs 66.06±56.36 moderate (minutes) 59.44±77.5 vs 59.23±83.77 Walk (minutes) 98.73±134.63 vs 75.59±79.29 sitting weekend (minutes) 358.27±216.67 vs 394.33±171.82 sitting weekday (minutes) 297.02±171.84 vs 339.41±176.8	Repeated measures ANOVA	Sex, self-reported BMI

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Table 2. Measures, Statistical Methods, and Outcome of the Studies Included in the Review (*continued*)

Author (year)	Physical activity outcome measure(s)	Components of arms	Intervention effect on physical activity outcome	Statistical analysis	Controls for baseline characteristics
Escaravajal-Rodriguez (2018) ³¹	Walking (% of participants)	1	PRE: Never: 37.9% 1 day/week: 14% 2 days/week: 15.1% 3 days/week: 9.5% 4 days/week: 5.6% 5 days/week: 6.7% 6 days/week: 3.4% 7 days/week: 7.8% POST: Never: 0 1 day/week: 6.2% 2 days/week: 10.2% 3 days/week: 14% 4 days/week: 14.1% 5 days/week: 20% 6 days/week: 11% 7 days/week: 24.5%	χ^2 test	Age, sex
Fontaine (2018) ³²	Sedentary behavior (minutes), light intensity physical activity (minutes), moderate intensity (minutes), vigorous intensity (minutes), steps (step/minute)	1	Sedentary: 4.9±3.0 minutes Light intensity: 6.1±3.9 Moderate intensity: 48.8±5.3 Vigorous intensity: 0.2±0.4 Total steps: 6,006.2±659.4 Steps: 100.1±9.5 steps/minute	Descriptive statistics	
Howe (2016) ³³	Daily steps (step/day)	2 Players, controls	Players: 4,257±2,698 (baseline) 45,123±3,371 (Week 1), 4,994±3,374 (Week 2), 4,693±3,104 (Week 3), 4,499±3,077 (Week 4), 4,108±2,927 (Week 5), 3,985±2,888 (Week 6) 955 increase (95% CI=697, 1,213) Controls: 4,127±2,930 (baseline)	Multivariate regression	Sex, age group, annual household income, race, bodyweight status, urbanity, and walkability index group, differences in time trends of daily steps

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Table 2. Measures, Statistical Methods, and Outcome of the Studies Included in the Review (*continued*)

Author (year)	Physical activity outcome measure(s)	Components of arms	Intervention effect on physical activity outcome	Statistical analysis	Controls for baseline characteristics
Kogan (2017) ³⁴	Walking, running, biking and skating (hours/day)	1	PRE: Walk: none group: 19±7.1; <1/2 hour group: 73±27.1; 1/2–1 hour group: 80±29.7; 1–2 hour group: 45±16.7; 2–3 hour group: 19±7.1; ≥3 hour group: 33±12.3 Bike: none 144±53.5; <1/2 hour 49±18.2; 1/2–1 hour 21±7.8; 1–2 hour 20±7.4; 2–3 hour 11±4.1; ≥3 hour 24±8.9 skate: none 196±72.9; <1/2 hour 16±5.9; 1/2–1 hour 16±5.9; 1–2 hour 12±4.5; 2–3 hour 6±2.2; ≥3 hour 23±8.6 Run: none 114±42.2; <1/2 hour 53±19.7; 1/2–1 hour 31±11.5; 1–2 hour 37±13.8; 2–3 hour 9±3.3; ≥3 hour 25±9.3 POST: Walk: none 6±2.2; <1/2 hour 35±13.0; 1/2–1 hour 57±21.2; 1–2 hour 60±23.3; 2–3 hour 54±20.1; ≥3 hour 57±21.2 Bike: none 131±48.7; <1/2 hour 41±15.2; 1/2–1 hour 30±11.2; 1–2 hour 21±7.8; 2–3 hour 16±5.9; ≥3 hour 30±11.2 skate: none 193±71.7; <1/2 hour 14±5.2; 1/2–1 hour 12±4.5; 1–2 hour 11±4.1; 2–3 hour 10±3.7; ≥3 hour 29±10.8 Run: none 87±32.3; <1/2 hour 46±17.1; 1/2–1 hour 46±17.1; 1–2 hour 36±13.4; 2–3 hour 22±8.2; ≥3 hour 32±11.9	Wilcoxon signed rank test, χ^2	
Liu (2017) ³⁵	Exercise frequency (days per week) and duration (minutes), daily steps (step/day)	1	PRE: Frequency: 2.8±1.6 Duration: 39.75±26.6 POST: Added frequency: 2.8±1.6 Added hours: 2.8±2.4 Daily steps: 1,600 steps increase; 5.6 miles increase	2 sampled t-test, Fisher's exact	

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Table 2. Measures, Statistical Methods, and Outcome of the Studies Included in the Review (*continued*)

Author (year)	Physical activity outcome measure(s)	Components of arms	Intervention effect on physical activity outcome	Statistical analysis	Controls for baseline characteristics
Ma (2018) ³⁶	Walking, running (km)	1	Walking and running: Increase 1,200 steps (0.96 km)	Averages, bootstrap, multilevel modeling	Weather, sex, age, education, income, BMI
Madrigal (2018) ³⁷	Walking (km)	1	Walking 24.7 (12.5–36.9) km	Multiple linear regression	Sex, age, education, SES, residence zone
Marquet (2018) ¹⁵	Daily steps (step/day)	2 Players, controls	Players: Baseline: weekdays: 7,626.6±3,847.5; weekends: 5,895.3±3,933.4 Controls: Baseline: weekday: 8,148.7±4,555.1; Weekends: 8,715.6±5,280.1; 1,526±599.140 step increase on weekends (active playing)	ANOVA, ANCOVA	
Militello (2018) ³⁹	Strenuous, moderate, mild physical activity (minutes/week; minutes/day)	1	PRE: Mild (minutes/day): 21.13±11.72 Mild (minutes/week): 79.59±68.06 moderate (minutes/day): 20.92 ±14.11 moderate (minutes/week): 60.45±62.82 strenuous (minutes/day): 16.57±15.49 strenuous (minutes/week): 48.5±60.69 POST: Mild (minutes/day): 26.39±11.38 Mild (minutes/week): 102.95±73.5 moderate (minutes/day): 25.11±13.03 moderate (minutes/week): 82.21±72.91 strenuous (minutes/day): 18.57±15.44 strenuous (minutes/week): 56±66.17	Pairwise t-tests, Bonferroni correction	
Nigg (2017) ⁴⁰	Strenuous, moderate, mild physical activity (minutes/day); sedentary time (minutes/day)	1	MVPA: increase 50 minutes/day Sedentary time: decrease 30 minutes/day	Bonferroni correction	Sex, race/ethnicity, BMI
Ruiz-Ariza (2018) ¹⁶	Moderate to vigorous activity MVPA (day/week); Walking (minutes/day, miles)	2 Players, controls	Players: PRE: 3.37±1.65 (MVPA) POST: 39.22±28.60 minutes/day walking; 33.7±20.15 miles Controls: 3.74±1.49 (MVPA)	Student's t-test, χ^2 , Spearman's correlation, ANOVA	Age, sex, SES, maternal education, BMI, and MVPA

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Table 2. Measures, Statistical Methods, and Outcome of the Studies Included in the Review (*continued*)

Author (year)	Physical activity outcome measure(s)	Components of arms	Intervention effect on physical activity outcome	Statistical analysis	Controls for baseline characteristics
Wattanapisit (2018) ³⁸	Sedentary time (minutes/week), energy expenditure, moderate and vigorous physical activity (MET minutes/week)	1	Phase 0 (baseline): MET minutes/week: 600 Phase 1: MET minutes/week: 996 Phase 2: MET minutes/week: 960 Phase 0: median sedentary time: 480 minutes/week Phase 1: median sedentary time: 270 minutes/week Phase 2: median sedentary time 300 minutes/week	Wilcoxon signed rank test, Friedman's test, Spearman's rank correlation	Sex, age, weight status, allowance, place of living, underlying illness
Wong (2017) ²⁹	Walking, jogging (minutes/week); MET (minutes/week); vigorous MET (minutes/week)	3 Players; ex-players, controls	Walking/jogging (minutes/week) Players: 98.43±151.21 minutes/week Walking MET (minutes/week): Players: 1,437.02±1,959.75 Moderate MET (minutes/week) Players: 443.66±1,184.71 Vigorous MET (minutes/week): Players: 678.04±1,313.33 Total MET (minutes/week): Players: 2,508.83±2,947.37 Walking/jogging (minutes/week) Former players: 68.23±141.82 minutes/week Walking MET (minutes/week): Ex-players and nonplayers: 1,346.61±1,740.48 Moderate MET (minutes/week) Ex-players and non/players: 433.52±1,178.05 Vigorous MET (minutes/week): Ex-players and non/players: 866.95±1,750.31 Total MET (minutes/week): Ex-players and nonplayers: 2,549.72±3,097.30	χ^2 , Wilcoxon signed rank test, univariate analysis, logistic regression	
Xian (2017) ³	Daily steps (step/day)	1	PRE: 5,678±2,833 POST: 7,654±3,616 1,976 steps increase	Paired t-test, Wilcoxon signed-rank test	

MVPA, moderate-to-vigorous physical activity.

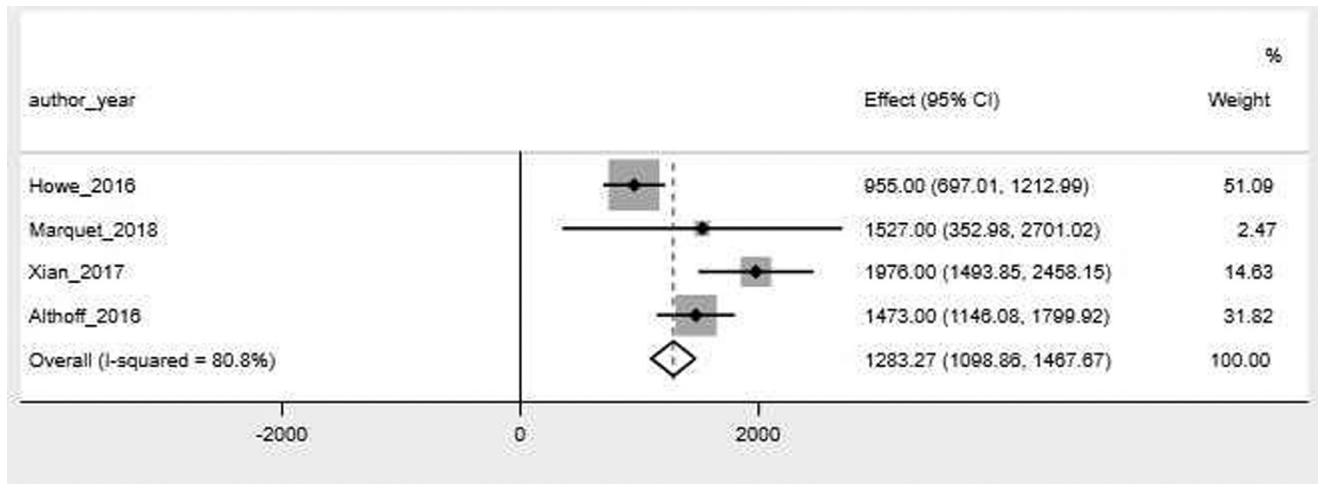


Figure 2. Forest plot.

to be emphasized and adequately protected during playing Pokémon Go.

To the authors' knowledge, this is the first systematic review that examined the influence of Pokémon Go on promoting physical activity. The heterogeneity in physical activity measures and outcomes made comparisons across studies difficult or infeasible. Studies included in the review also differed on the duration of play and comparison (i.e., pre- versus post-play or players versus controls). All except 2 studies evaluated the effect on adults, which raises the question of generalizability of study findings to children and adolescents.¹⁶ Moreover, several studies did not mention the reliability and validity of the instruments used to measure outcomes. Most studies adopted an observational study design (cross-sectional or cohort studies), limiting the ability to infer a causal relationship linking Pokémon Go to physical activity. In fact, considering the rapid popularity of the game, it would be almost impossible to conduct a prospective study. In addition, it is essential to note that there is a vast array of implications that go along with the game beyond physical activity. Factors such as neighborhood exploration, playing outside, socializing with peers, intergenerational play, and play with unknown people are important dimensions of the Pokémon Go experience that should be considered as well when evaluating the overall effectiveness of the game.

Future research should not only examine the long-term effects of Pokémon Go to understand its sustainability in physical activity promotion, but also consider other augmented reality games, particularly in further meta-analyses. On one hand, RCTs with a larger sample size and longer follow-up period are warranted to demonstrate causality. On the other, though, there are serious potential challenges such as users' attrition rate

because of low engagement with Pokémon Go, particularly over time. As the novelty of this specific game, not the technology overall, tends to wear off with time, accelerometer and other objective physical activity measures could be useful to accurately record frequency, intensity, and duration of physical activity. One of the studies included in the review found that after 3 months, more than 80% of Pokémon Go players had discontinued play.³⁰ Likewise, another study indicated that after 2 months of play, 23% of the study participants planned to stop playing or had already stopped playing Pokémon Go.³⁵ Current, although limited, research on the impact of Pokémon Go on children and families has reported that spending time together and presence of in-app events were the most common reasons for gameplay for parents and children and re-engagement with the game, respectively.³⁹ It is important to note that although daily steps increased significantly during the first week of the game installation, this association had attenuated by the 6th week,³³ further raising concerns on the long-term health impact of Pokémon Go. Additional psychosocial assessment may provide valuable information on what contributes to a sustainable game play.

Limitations

This review has several limitations. Studies included in the review were mostly observational, and none adopted an RCT design. While conducting the search for this review, the researchers did find 2 ongoing RCTs that examined the impact of Pokémon Go on physical activity outcomes, but no results have been reported to date. The small number of studies may not have the statistical power to detect publication bias. Therefore, the results may not present a precise estimate of the effect size. Studies were heterogeneous in intervention duration

and exposure and outcome variable measures, making it difficult or sometimes infeasible to compare across and generalize from the included studies. The attrition rate among the included studies ranged from 0% to 56%, which questions the long-term sustainability of Pokémon Go as a physical activity intervention. The estimated pooled effect from the meta-analysis was solely based on adult participants. Therefore, the results may not be generalizable to children and adolescents, who are certainly one of the main Pokémon Go targeted players.

CONCLUSIONS

This study systematically reviewed the scientific literature linking the Pokémon Go mobile game to physical activity engagement. Playing Pokémon Go was found to be associated with a statistically significant but clinically modest increase in the number of daily steps taken among game players. Although this game was not developed explicitly as a mobile health-promoting application, the results of this review suggest that incorporating exercise into mobile games for entertainment may influence behavior change. One challenge for future physical activity intervention using Pokémon Go is to retain active engagement once the initial novelty wears off. Additional studies with longer follow-up periods and experimental study design are needed to assess whether and to what extent Pokémon Go can be used to promote physical activity at the population level for a sustained period.

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SUPPLEMENTAL MATERIAL

Supplemental materials associated with this article can be found in the online version at <https://doi.org/10.1016/j.amepre.2019.09.005>.

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